

C868 – Software Capstone Project Summary

Task 2 – Section A



Capstone Proposal Project

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Business Problem

The Customer

TechPro Solutions is a rapidly growing technology consulting firm that provides innovative solutions to streamline business operations. With offices in major cities across the US, Canada, and the UK, TechPro caters to a diverse clientele ranging from startups to Fortune 500 companies. The company employs a dynamic team of software engineers, project managers, and business analysts who collaborate to deliver cutting-edge solutions tailored to meet each client's unique needs. Their mission is to empower businesses with technology-driven solutions that enhance efficiency, productivity, and profitability. TechPro aims to expand its service offerings and establish itself as a leader in digital transformation solutions in various industries as part of its long-term goals. With a projected growth trajectory, TechPro anticipates enhancing internal operations to support its expanding client base and workforce.

Business Case

TechPro Solutions currently faces challenges in efficiently managing internal meetings and external client interactions. With a diverse range of meeting types and stakeholders, there needs to be standardized processes and tools to organize these collaborations. This lack of tools leads to scheduling conflicts, disjointed communication channels, and meeting preparation and follow-up inefficiencies. To address these challenges, TechPro Solutions requires a comprehensive customer and meeting management application that can centralize scheduling, facilitate the adding and removal of customers, and provide easily accessible meeting and customer data. The application will serve as a centralized

platform for organizing internal interactions, such as staff meetings, training sessions, project briefings, and external interactions, including client consultations and product presentations.

Fulfillment

The meeting management application developed for TechPro Solutions will be a user-friendly stand-alone application accessible to authorized personnel within the organization. It will contain a login screen that requires users to provide a username and password to log in. These security measures will ensure that only authenticated users can access sensitive customer and meeting information. The application will feature distinct sections for creating internal and external meetings, allowing users to select from predefined meeting types. Users can easily schedule meetings by selecting the type, date, time, and participants. The application will prevent double booking and provide upcoming meeting notifications to users upon login.

The application also includes a Customer creation page. It will have the functionality to create, edit, and delete customer profiles. Each customer is auto-assigned a distinct Identification number. Users can create customers with selected names, addresses, and phone numbers. These profiles can be associated with meetings to streamline meeting organization. The application will include a reports page to help find and organize meetings. It will include monthly meeting totals, meetings by contact, and meetings by type. The customer page will have a search bar that allows users to search by ID and Name. The meeting page will include radio buttons that allow filtering of meetings by the current week and month. It will also include a search bar to search by ID, Type, and Location.

We will provide user training and support to facilitate the seamless adoption of the new meeting management system. In addition, we will provide a user guide for reference and new-user training. Overall, the meeting management application will enable TechPro Solutions to streamline internal

operations, enhance client interactions, and drive organizational efficiency as it continues to grow and expand its service offerings in the competitive technology consulting market.

Existing Gaps

TechPro Solutions currently grapples with the challenges of coordinating internal meetings and client interactions using an outdated manual process. The existing method relies heavily on a shared paper calendar or planner to schedule and record meetings, presenting several notable organizational efficiency and communication deficiencies. Without a centralized system, accessing and referencing meeting details is cumbersome and fragmented, leading to confusion among team members. The lack of real-time visibility into meeting availability further compounds the problem, increasing the likelihood of scheduling conflicts and hindering effective coordination.

Currently, communication surrounding meeting scheduling and updates occurs through informal channels such as verbal exchanges or emails, lacking a structured platform for transparent and accountable communication. Consequently, errors, misinterpretations, and delays are common occurrences, impeding the smooth coordination of meetings and the sharing of vital information. The manual process results in meeting preparation and follow-up inefficiencies as team members need clear visibility into upcoming meetings to prioritize tasks. The reliance on manual paper-based methods for meeting management poses significant challenges for TechPro Solutions, highlighting the pressing need to develop a comprehensive meeting management application to address these deficiencies and enhance organizational productivity.

SDLC Methodology

For the development of the meeting management application, MeetUp, we have adopted the Agile methodology as our preferred SDLC approach. Agile methodology aligns well with the dynamic nature of software development, enabling us to deliver value incrementally while remaining adaptable to changes in requirements. This methodology emphasizes collaboration, flexibility, and iterative development, allowing us to respond effectively to feedback and evolving customer needs throughout the project lifecycle.

The Agile approach will guide our development process through the following key phases:

Requirements Gathering and Analysis:

We will collaborate closely with TechPro Solutions stakeholders to elicit and prioritize requirements for the meeting management application. We will ensure a comprehensive understanding of TechPro Solutions' needs and expectations through user stories and regular communication.

Iteration Planning:

Based on the prioritized requirements, we will organize our development work into iterative cycles, known as sprints. Each sprint will focus on delivering a set of functionalities that align with TechPro Solutions' immediate needs and goals. The duration of each sprint will be determined based on the complexity of the feature development and the team's capacity.

Development:

During each sprint, our cross-functional development team will work collaboratively to design, develop, and test the features identified for that iteration. Continuous integration and frequent testing will ensure that each application increment is functional, stable, and aligned with TechPro Solutions' requirements.

Review and Feedback:

At the end of each sprint, we will conduct a review meeting with TechPro Solutions stakeholders to demonstrate the newly developed functionality and gather feedback. We will use this feedback to refine and prioritize the product backlog for subsequent sprints, ensuring we remain responsive to TechPro Solutions' evolving needs.

Deployment and Iterative Improvement:

As TechPro Solutions completes and validates each increment of the application, it will be deployed to a staging environment for further testing and evaluation. Any necessary adjustments or enhancements identified during this phase will be incorporated into future sprints, allowing us to improve the application continuously over time.

By embracing the Agile methodology, we aim to deliver a high-quality meeting management application that meets TechPro Solutions' needs, enhances operational efficiency, and provides value from the earliest stages of development. Our iterative approach will enable us to adapt to changing requirements, mitigate risks, and deliver a solution that aligns with TechPro Solutions' vision for MeetUp.

Deliverables

As mentioned earlier, Agile software development emphasizes iterative and collaborative approaches. In this methodology, each project lifecycle stage yields specific deliverables tailored to the Agile framework. These deliverables will be categorized into project deliverables, overseen by the Project Manager, and product deliverables, which encompass the tangible software product provided to the customer. The subsequent section will elaborate on these deliverables.

Project Deliverables

- Requirements List
 - During the brainstorming phase, we will meet with the customer to gather a comprehensive list of application requirements.
- Functionality Change List
 - Documentation outlining changes and needed additions to the existing software from C195
- Timeline
 - Will delineate critical milestones and deadlines, facilitating effective project management and coordination
 - It helps stakeholders understand the sequence of activities and dependencies.
- Code Repository
 - Establishment of a code repository to store application code
 - Allows for rollback and version control
- Database ERD

- Entity-relationship diagram (ERD) depicting the database schema for the application
 - Helps stakeholders conceptualize how the software will work
- Unit Testing Plan
 - An in-depth plan outlining unit and functional testing procedures, created during the Testing Plan phase
- Maintenance Agreement
 - Agreement outlining the terms and responsibilities for ongoing maintenance of the application

Product Deliverables

- Mid-Fidelity Wireframe
 - A mid-fidelity wireframe representing the planned changes to the application's user interface
- Functional Application
 - The functional application implements changes from the planning and design phases according to the requirements gathered during the Brainstorming phase.

- User Guide
 - In-depth documentation providing instructions on how to utilize the application.
- SQL Scripts for Database Creation
 - Necessary SQL scripts for database creation and application deployment
 - Includes DDL and DML scripts
- Customer Feedback from Acceptance Testing
 - We will gather feedback from the customer based on a series of tests to verify that the application meets the requirements during the Testing Phase.

Implementation

The project manager will coordinate the implementation of the MeetUp meeting management application, oversee various teams, and ensure seamless integration of the software into the production environment. The implementation will involve several steps, including preparing the necessary resources for application deployment. It also includes finalizing SQL scripts for database creation and packaging the application files into a zip file for distribution to TechPro Solutions. The Project Manager will work closely with the development team to ensure all components are correctly configured and ready for deployment.

After preparing the resources, the deployment process will commence, with the Project Manager overseeing the distribution of SQL scripts and application files to TechPro Solutions. We will stage the deployment to minimize disruptions to TechPro's operations, with clear communication channels established to inform users of the upcoming changes. Web Administrators will be responsible for configuring the web server environment and ensuring that the application is accessible to authorized personnel within the organization. The Project Manager and development team will promptly address any potential issues or concerns that arise during deployment to ensure a smooth transition.

Validation and Verification

The application incorporates a login screen as a fundamental security measure, necessitating valid user credentials for access. This login requirement serves as a gatekeeper, ensuring that only authorized personnel within TechPro Solutions can enter the application environment. The application guards against unauthorized access through username and password authentication, safeguarding sensitive customers and meeting information from potential breaches.

Moreover, the application's functionality prevents the creation of overlapping meetings, a critical feature to streamline scheduling and avoid conflicts. The application maintains data integrity and coherence by enforcing this constraint, ensuring that meeting schedules remain accurate and manageable. This capability is integral to enhancing organizational efficiency, as it mitigates the risk of scheduling errors and optimizes resource utilization.

While the application does not introduce additional features beyond these essential functionalities, the primary focus remains verifying user input integrity to uphold system security and data accuracy. Comprehensive testing procedures, including unit and acceptance testing, will be used to

validate the application's compliance with TechPro Solutions' requirements and expectations. Overall, the validation and verification process will involve collaboration between the development team, TechPro Solutions stakeholders, and end-users to ensure that the MeetUp meeting management application delivers the desired functionality and effectively meets the needs of its users.

Environments and Costs

Programming Environment

The meeting management application for TechPro Solutions will require a specific hardware and software environment to ensure its successful development and deployment. The hardware requirements include standard desktop or laptop computers for the development team, with sufficient processing power and memory to support Java development tools and MySQL database management. The software environment will consist of the following components: Java Development Kit (JDK), which provides the necessary tools and libraries for Java development, including compilers, debuggers, and runtime environments. The development team will use an Integrated Development Environment (IDE), such as IntelliJ IDEA or Eclipse, to write, debug, and test code efficiently. MySQL Database will serve as the backend database management system for storing and managing meeting data, and it will be responsible for storing meeting schedules, participant information, and other relevant data. A web server, such as Apache Tomcat, will host the Java application and serve it to users over the internet or intranet.

Environment Costs

The costs associated with the software application environment for the meeting management system are relatively minimal and consist primarily of startup and recurring fees. We plan to host the environment in a shared environment, distributing costs among multiple organizations. Database Maintenance incurs a nominal fee of \$500 per year for maintaining the MySQL database, covering unlimited storage size and ensuring 99.8% uptime for reliability and scalability of the application's data storage needs. Web Server Maintenance costs \$300 per year, encompassing ongoing maintenance and upgrades for the Windows Server, Internet Information Services (IIS), and any additional components such as ColdFusion. Client Access Fees vary depending on the number of devices accessing the application, with an annual fee of \$40 per device covering operating system and network upgrades for each client device connected to the network. These costs contribute to the overall operational expenses of the meeting management application, ensuring the software environment's reliability, security, and performance.

Human Resource Requirements

The labor required to complete the application will involve a team of skilled professionals, including a project manager, software developers, a designer, and a quality assurance (QA) specialist. The developers will consume the largest share of human resources, accounting for approximately 75% of the total hours and costs associated with the project. The project manager will oversee the project's execution, ensuring the accomplishment of significant milestones and efficient allocation of resources. The designer will focus on creating an intuitive and visually appealing user interface, while the QA specialist will conduct rigorous testing to ensure the application's quality and reliability. In summary, the human resource requirements for the meeting management application will involve a dedicated team working collaboratively to deliver a high-quality, user-friendly software solution within the specified timeline and budget constraints.

Resource	Rate * Time	Total
Project Manager	\$60/h * 95h	\$5700
Designer	\$45/h * 40h	\$1800
Software Developers	\$65/h * 110h * 2	\$14,300
QA Specialist	\$40/h * 30h	\$1200
TOTAL Human Resource Cost		\$23,000

Project Timeline

Phase	Milestone/Task	Deliverable	Description	Dates
Brainstorming	Task 1/ Requirements for application	Requirements List	The project manager will meet with the customer to determine the requirements	4/1/2024 – 4/3/2024
Planning	Determine Application Changes	Functionality Change List, Timeline	Determine changes/additions to existing	4/4/2024 – 4/7/2024

			software, Create a timeline	
Planning	Establish a Foundation for the Project	Outline, Code Repository	Project Managers will create a documentation outline and repository to store application code	4/5/2024 – 4/9/2024
Application Design	Review and Change Application UI	Mid-Fidelity Wireframe	The designer will analyze the current application design and adjust it according to the requirements	4/8/2024 – 4/11/2024
Database Design	Create Database Schema For Application	Database ERD	Developers will create SQL scripts to establish tables with appropriate relationships and data	4/12/2024 – 4/13/2024
Testing Plan	Create Testing Plan	Unit Testing Plan	The project manager and developers will create an in-depth plan for unit and functional testing	4/13/2024 – 4/15/2024
Implementation	Create a Functional Application	Functional Application	Developers will implement changes from the planning and design phases according to the requirements from the Brainstorming phase.	4/16/2024 – 4/24/2024

User Documentation	Create User Guide	User Guide	The developers and project managers will create an in-depth User Guide with instructions on how to utilize the application	4/18/2024 – 4/23/2024
Testing Phase	Execute Functional and Unit Tests	Unit Test Results	QA specialists will create and run functional tests against the initial requirements. Execute and document various Unit Tests.	4/25/2024 – 4/27/2024
Testing Phase	Execute Acceptance Testing	Customer Feedback from Acceptance Testing	Customers will run tests to verify that they meet the requirements.	4/28/2024 – 4/30/2024
Deployment	Prepare Necessary Resources for Application Deployment	SQL Scripts for Database Creation, Application zip File	The project manager will provide customers with SQL scripts, application zip files, and the User Guide.	5/1/2024 – 5/2/2024
Maintenance	Create Maintenance Plan	Maintenance Agreement	The project manager will meet with the customer to create a maintenance agreement for the application	5/6/2024